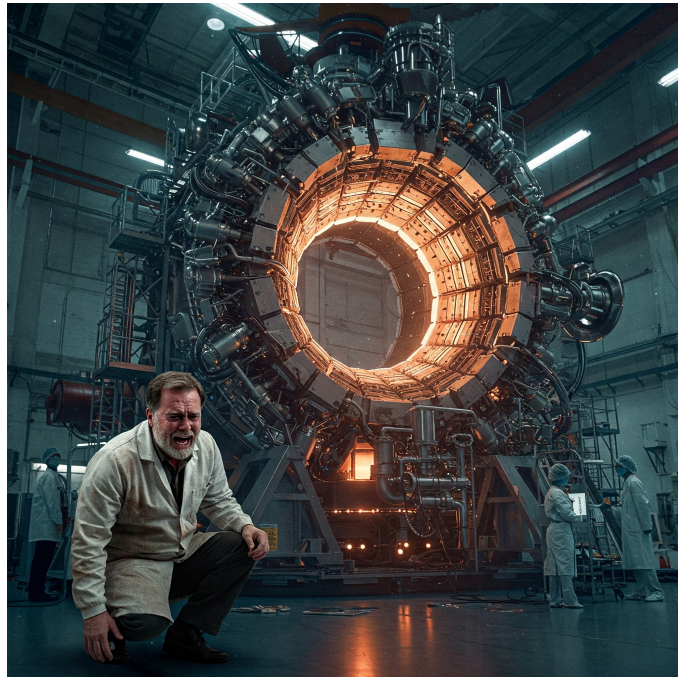

Recap Magnetic Confinement and MHD of Fusion Plasmas

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April 14, 2025



¹Figure 0: a picture of you while preparing for this exam.

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Preface

Lecture 1: Burn criteria

Collisions

² For fusion to happen the two nuclei need to hit each other. The cross-section σ is the imaginary (capture) area particles have for which they are susceptible for collisions.

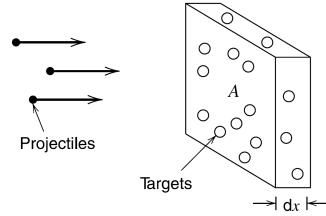


Figure 1: Concept behind cross section.

We define the probability and the mean free path

$$prob = \frac{\sigma}{A} N = \sigma n dx, \quad \boxed{\lambda_{mfp} = \frac{1}{\sigma n}} \quad (1)$$

,with n the density (of the target), and the mean time between collisions and associated collision frequency

$$\tau_c = \frac{\lambda_{mfp}}{v}, \quad \boxed{f_c = \frac{1}{\tau} = n\sigma v} \quad (2)$$

where one should keep in mind that σ can be dependent on velocity (and thus indirectly temperature). One simple model to determine the cross section is the hard sphere cross-section where you very simple approximate both particles as hard spheres. In reality other effect such as quantum mechanics should be taken into account.

Thus the power density is

$$\boxed{S_f = E_f n_1 f_c = E_f n_1 n_2 \sigma v} \quad (3)$$

where E_f is the fusion power per collision. The multiplication by n_1 is needed to account for the density of the projectiles.

There are multiple ways to model the cross-section (see figure 2a):

- The hard-sphere model is the simplest and imagines the particles to be spheres regardless of speed or directness of the collision.
- The classical model takes into account the minimum energy/speed needed to overcome the Coulomb barrier.

And finally there is the quantum model, which takes into account tunneling

$$T(E_{kin}) = e^{\sqrt{-\frac{E_G}{E_{kin}}}} \quad (4)$$

where E_G is the Gamow factor. This has different results at different energies:

- Low energies: not enough energy overcome Coulomb barrier, may tunnel.
- Gamow peak: enough energy to tunnel and enough interaction time, can also overcome barrier.
- High energies: not enough interaction time.

²Based on the slides 6-10 and Freidberg chapter 3.1-3.4

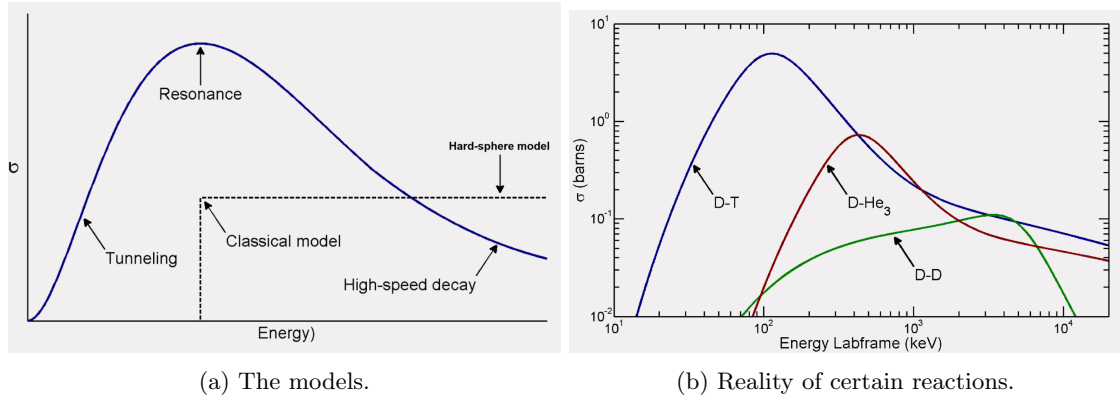


Figure 2: Cross-sections and their models.

Coulomb collisions/Bremsstrahlung

³ A big problem is Coulomb collisions where particles ‘bounce’ of each other (see figure 3). Here a lot of the energy is lost. It can be model based on a simple impact parameter b

$$\frac{2\Delta E_{co,90}}{1 + (b/b_{90})^2} = \Delta E_{co} \quad (5)$$

where b_{90} and $E_{co,90}$ are the impact parameter and the loss per collision in the case of a 90 degree bounce.

When calculating the total effect the impact parameter it typically integrated until the Debye length. This results in the Coulomb logarithm

$$\ln(\Lambda) = \ln\left(\frac{\lambda_D}{b_{90}}\right) \approx 15 \quad (6)$$

which when then taking into account all angles result in a very high cross-section

$$\sigma_{co} = 4 \ln(\Lambda) \sigma_{co,90} \quad (7)$$

These result in the proportionalities

$$b_{90} \propto \frac{1}{v^2} \rightarrow \Delta E_{co} \propto v^2, \quad \sigma_c \propto \frac{1}{E_{kin}^2} \propto v^{-4} \quad (8)$$

This means the total power loss per particle is proportional to nv^{-1} . But due to the Coulomb logarithm and the decay of the cross section at higher temperatures this is very hard to get below the fusion power, assuming no confinement.

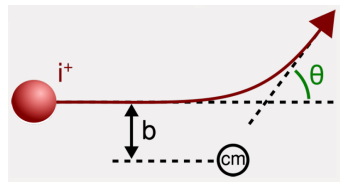


Figure 3: Concept behind Coulomb collisions.

³Based on the slides 11-23. See also Freidberg chapter 9.1-9.3

Fusion power

⁴ The reaction rate (per volume) is

$$R_{12} = n_1 n_2 \sigma v \quad (9)$$

where we can use a velocity distribution function to eliminate v . This results in a integral of two distribution function of which one is significantly lower and thus results in a peak in the reaction rate graph (see figure 4). Here D-T is clearly the best and has a peak around $T \approx 10 - 100$ KeV.

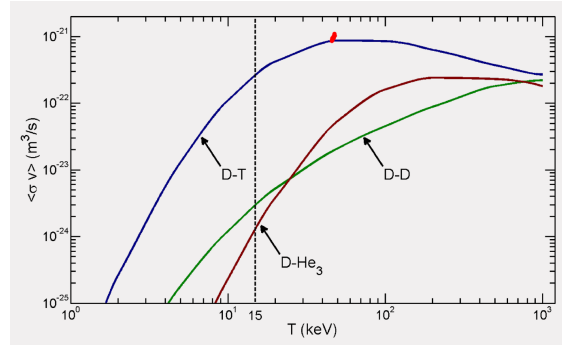


Figure 4: Reaction rates.

Ik snap
slide 27
totaal niet

Energy balance

⁵ We can then make an energy balance

$$\frac{dW}{dt} = P_\alpha + P_{ext} - P_{rad} - P_\kappa \quad (10)$$

where P_α is the heating due to alpha particles from fusion $\approx 1/5 P_{fus}$ (due to the ratio of energy between the alpha particle and the neutron), P_{ext} is the external heating, P_κ is the transport losses and P_{rad} are the radiation losses. Of which Bremsstrahlung, caused by Coulomb collisions, is the largest one. Thus the components are

$$P_\alpha = E_\alpha \frac{n^2}{4} \langle \sigma v \rangle, \quad P_{ext} = \frac{P_{fus}}{Q} \approx \frac{5P_\alpha}{Q}, \quad P_{rad} \approx P_{brehm} \propto Z^2 n_e^2 \sqrt{T}, \quad P_\kappa = \frac{W}{\tau_E} \quad (11)$$

where τ_E is the (transport) energy confinement time, Q is the power gain, and W is the confined energy

$$W = \frac{3}{2} n k_b T \quad (12)$$

There are two definitions of energy confinement time, one based on transport P_κ (as before) and one based on all losses

$$P_{loss} = P_{rad} + P_\kappa = \frac{W}{\tau_E^*} \quad (13)$$

since transport can be difficult to predict empirical scaling laws are often used for τ_E^* . If this is plugged into energy balance we get

$$\frac{dW}{dt} = P_\alpha \left(1 + \frac{5}{Q}\right) - P_{loss} \quad (14)$$

if you then calculate the Q of the steady state you can determine if the reactor is:

⁴Based on the slides 24-30. See also Freidberg chapter 3.1-3.4

⁵Based on the slides 31-49. See also Freidberg chapter 4.1-4.8

Uitleggen
dat factor
4 komt
omdat
 $n^2/4 =$
 $n_e n_i =$
 $n/2 \cdot n/2$

Maybe re-
move or
re-add the
factor $\frac{1}{2}$
or spec-
ify that
 $p = nk_b T$
and $W =$
 $\frac{3}{2} p dx^3$

- Break-even (equal output power to losses) $Q = 1$
- Ignition (alpha power compensates for losses) $Q = \infty$

For simplifications the calculation can be done by only assuming Bremsstrahlung. The results can be seen in figure 5 where the (ideal) triple product is laid out (for different temperatures). The radiation causes ignition to not be possible below $T \approx 4.4 \text{ keV}$.

Additionally you have radiation limit⁶ which stems from the fact that

$$P_{brem} \approx P_{rad} \leq P_{loss} \quad (15)$$

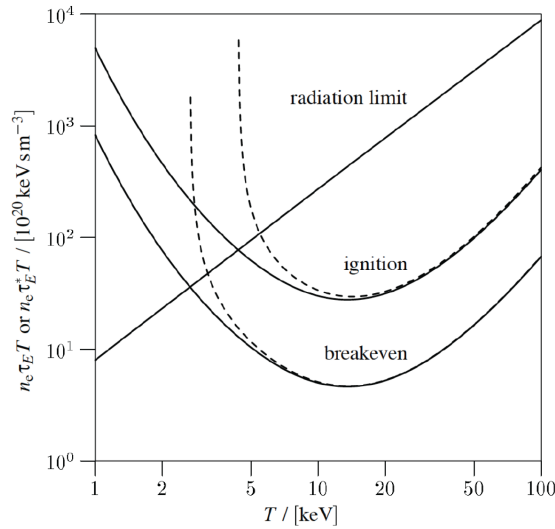


Figure 5: The triple product required for ideal break-even and ignition. The dashed lines include radiation losses.

Berekening van slide 36 hier neer zetten

Example: Tripple product calculation (exam February 2024 problem 6)

Derive the fusion triple product $nT\tau_E$ for the ignition criterion for D-T fusion. Make here a difference between conduction losses and radiation losses. Use for the radiated power density of a D-T plasma: $p_{rad} = C_b n_e^2 \sqrt{T}$

Assume $T_e = T_i = T$, and $n_D = n_T = 0.5n_e$.

(NB: Note that the right hand side of your equation should not contain the density or energy confinement time, only the temperature as parameter)

4.0 points · Open · 1/2 Page

Model answer

Ignition criterion:

$$P_\alpha = P_{cond} + P_{rad}$$

This translates into the following equation for the power densities

$$\frac{1}{4}n^2 \langle \sigma v \rangle E_\alpha = \frac{3nT}{\tau_E} + C_b n^2 \sqrt{T}$$

Therefor the triple product gets:

$$nT\tau_E = \frac{12T^2}{\langle \sigma v \rangle E_\alpha - 4C_b T^{3/2}}$$

⁶<https://physics.stackexchange.com/questions/331393/how-to-calculate-the-bremsstrahlung-limit-in-the-fusion-triple-p>

Explain why no factor $\frac{k_b}{2}$ at confinement energy

The resulting Helium in the plasma has two effects

- Decreases DT density $\rightarrow$ lower fusion power.
- Increases radiation losses.

these can be calculated by using fraction and the superposition's of equation 11. The ratio of the confinement time of the helium and energy is

$$\rho = \frac{\tau_{He}}{\tau_E} \quad (16)$$

which must be below 10 or else the losses dominate (see figure 6). The exact power reduction is

$$\frac{1 - P_{polute}}{P_{pure}} = (1 - 2f_{He})^2 \text{ with } f_{He} = \frac{n_{he}}{n_e} \quad (17)$$

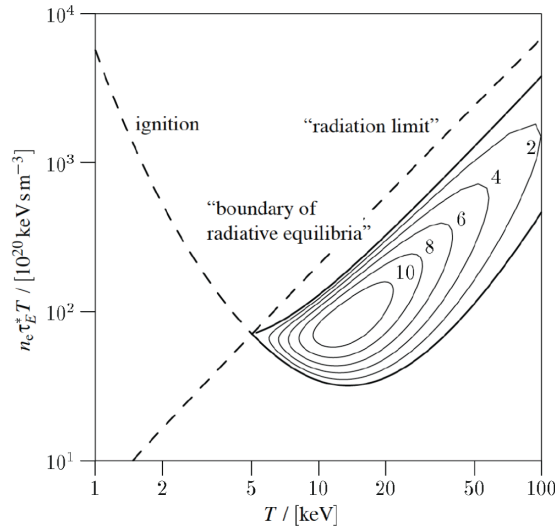


Figure 6: The triple product curve with curves for different values of helium polution ρ .

Example: Effect of Helium (exam April 2024 problem 7 and April 2019 problem 1.a.2)

[4 pt] Calculate the reduction of the fusion power for a burning D-T plasma with temperature $k_B T = 20 \text{ keV}$ when the ratio of helium confinement time to energy confinement time $\rho = \tau_{He}/\tau_E$ increases from $\rho = 0$ to $\rho = 10$. Neglect the radiation losses due to the bremsstrahlung for this case. Hint: write down the equation for the burn condition and the equation for the definition of the helium particle confinement time.

Solution: define $f = n_{He}/n_e$, then the power reduction factor is $(1 - 2f)^2$. f can be found by eliminating τ_E from the burn condition and from the definition of the helium confinement time $\tau_{He} = n_{He}/(dn_{He}/dt) = \rho \tau_E$. This yield as equation for f : $f = 3/(E_\alpha/(T\rho) + 3/2) = 3/19$. Thus the reduction in power is $(13/19)^2 = 47\%$.

Clarrrify how the definition of helium confinement time and burn condition lead to f_{He} .
See slide 41

$$P_\alpha = \frac{1}{4} f_i^2 n_e^2 \langle \sigma v \rangle E_\alpha = P_k = \frac{3(1+f_i+f_{He}+f_z)n_e T}{2\tau_E}$$

$$f_z = 0$$

$$f_i = 1 - 2f_{He}$$

$$P_\alpha = \frac{1}{4} (1 - 2f_{He})^2 n_e^2 \langle \sigma v \rangle E_\alpha = P_k = \frac{3(2-f_{He})n_e T}{2\tau_E}$$

If one then again includes all terms from equation 10 a figure like 7. From that we can conclude that we must heat the plasma to $T = 7 \text{ keV}$ after which it will heat itself and reach a stable equilibrium at $T = 16 \text{ keV}$.

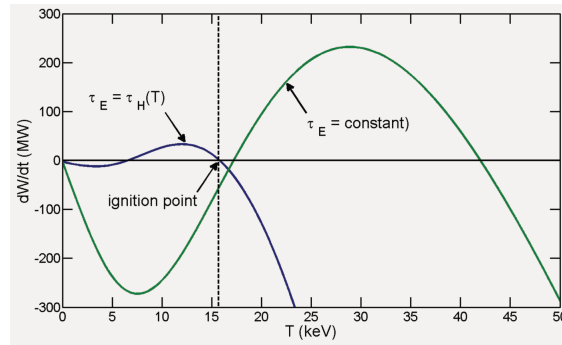


Figure 7: A graph of $\frac{dW}{dt}$.

The result is very similar to the Lawsons criteria which differs in that it also includes an conversion efficiency to power.

Slide 47 over andere limieten

Lecture 2: Orbits

Gyration

7

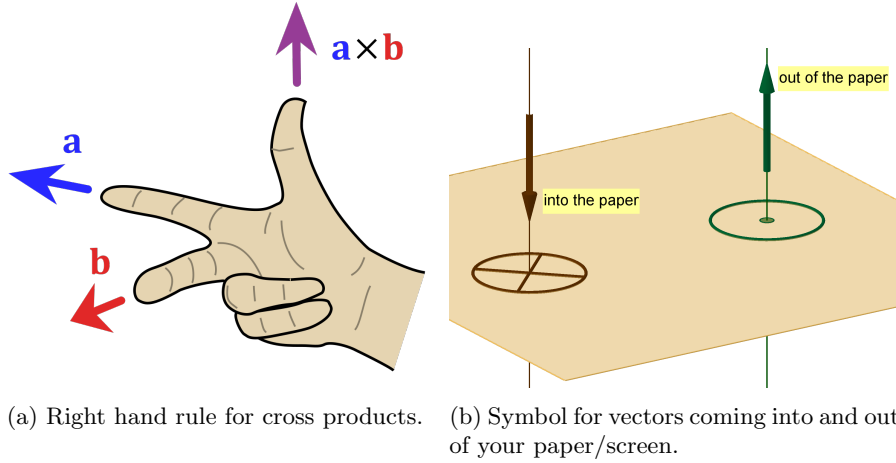


Figure 8: Things to remember about vectors.

The particles in a plasma behave according to the Lorentz force

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad (18)$$

refer to figure 8 for the direction. The cross product with velocity causes a gyration parallel to the magnetic field (clockwise for electrons, see figure 9). The center of the gyration/particle motion is called the guiding center on which physics can be done separately.

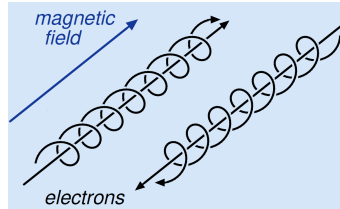


Figure 9: A example of magnetic gyration.

The gyration frequency/electron cyclotron frequency is

$$\omega_{ce} = \frac{qB}{m}, \quad f_{ce}[\text{GHz}] \approx 28[\text{GHz/T}] \cdot B[\text{T}] \quad (19)$$

and the Larmor radius is

$$\rho_c = \frac{m_e v_{\perp}}{eB} \quad \text{with} \quad mv_{\perp} = \sqrt{2mk_b T} \quad (20)$$

these can be derived from a simple centrifugal force calculation

(Electric) force drift (E cross B)

⁸ A force on the particle perpendicular to the B field can cause a drift. That is caused by the fact that the particle changes velocity during its gyration (slower at the higher potential) and

⁷Based on the slide 4. See also Freidberg chapter 8.1-8.3

⁸Based on the slide 7-12. See also Freidberg chapter 8.4

thus experience higher magnetic forces in one direction

$$\vec{v}_{F \times B} = \frac{\vec{F} \times \vec{B}}{qB^2} \quad (21)$$

most commonly a electric field, which causes a E X B drift. This kind of drift is equal for both ions and electrons since the charge chances out. While, for example gravitational drift, is in opposite directions.

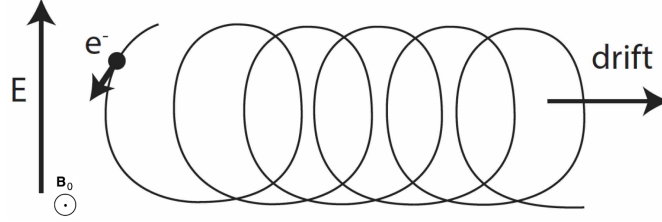


Figure 10: Example of $E \times B$ drift. In this case the magnetic field comes out of the paper.

Grad B drift

⁹ A gradient in the B-field has a similar effect. When in the part with stronger B-field the force will be stronger the radius becomes smaller while for the smaller B-field the radius becomes larger. Thus resulting in a grad B drift

$$\vec{v}_{\nabla B} = -\frac{mv_{\perp}^2}{2qB^3}(\nabla B) \times \vec{B} \quad (22)$$

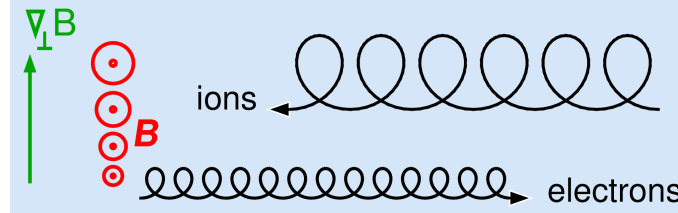


Figure 11: Example of ∇B drift. In this case the magnetic field comes out of the paper.

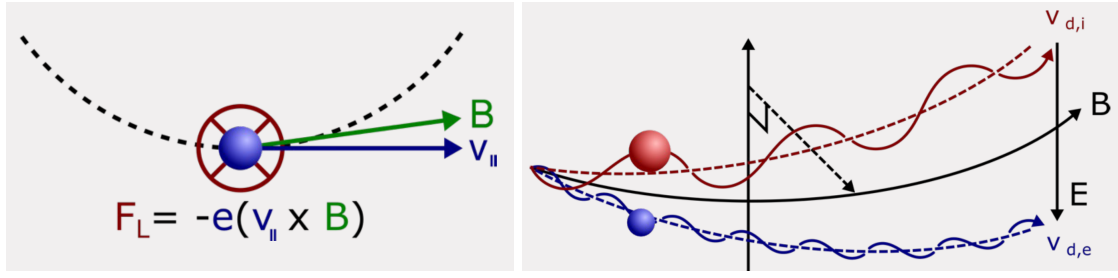
Curvature drift

In a curved magnetic field the velocity and magnetic field might not allign. Thus (according to the Lorentz force) causing a force perpendicular to the curvature and magnetic field. Note that the direction of the curvature B drift is (contrary to expectations) in opposite directions for ions and electrons.

⁹Based on the slide 13 and lecture 6 slide 12. See also Freidberg chapter 8.5-8.6

Consider adding generalized formula 8.53 from book

Expand



(a) Curvature B drift of an electron. In this case the force goes into the paper.

(b) Side view of curvature B drift.

Figure 12: Curvature B drift.

Add polarization drift, <https://gemini.google.com/app/ea97015a304c4acf>

Drift in a Tokamak

¹⁰ Linear confinement via a mirror wouldn't work due to end-losses but a simple toroidal B-field is also not enough for confinement due to drifts

- There is a gradient and curvature in B and thus a grad B drift which is up for ions and down for electrons.
- That causes a electric field which then causes an $E \times B$ drift outwards.

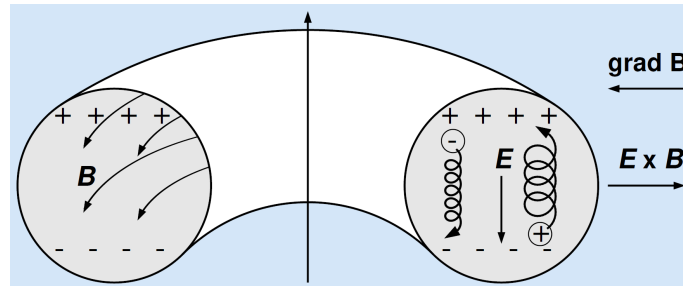


Figure 13: The drift effects in a toroidal system.

To solve these drifts a helically twisted field is needed and thus:

- Stellarators use complex external coils.
- Tokamaks use helical magnetic fields (requires/causes a plasma current) which “short-circuit” the top and bottom of the plasma.

Mirror

¹¹ The first adiabatic invariant (something that doesn't change in the short term) is the magnetic moment which means that in the order of the gyrations the angular momentum stays the same. With energy conservation this results in

$$\mu = \frac{mv_{\perp}^2}{2B} = \frac{q\omega_{ce}}{2}\rho^2 = \text{Const} \rightarrow E_{kin} = \frac{1}{2}mv_{\perp}^2 + \frac{1}{2}mv_{\parallel}^2 = \boxed{\frac{1}{2}mv_{\parallel}^2 + \mu B = \text{Const}} \quad (23)$$

¹⁰Slide 23

¹¹Based on the slide 6, 16-22. See also Freidberg chapter 8.7-8.9

which means. Increasing B means (higher $v_{\perp}$ and) thus lower $v_{\parallel}$. Thus if magnetic field is high enough (such $v_{\parallel} \leq 0$) that a particle can be reflected/trapped in a sort of magnetic mirror (see figure 14a). The trapping condition is

$$\frac{v_{\parallel,0}}{v_{\perp,0}} \leq \sqrt{\frac{B_{max}}{B_{min}}} - 1 \quad (24)$$

where $v_{\parallel,0}$ and $v_{\perp,0}$ are the velocities at B_{min} . If this condition is not satisfied it is a passing particle. By plotting the parallel and perpendicular velocity on a unit sphere and applying the condition one can see what the loss cone is and what the trapping fraction is.

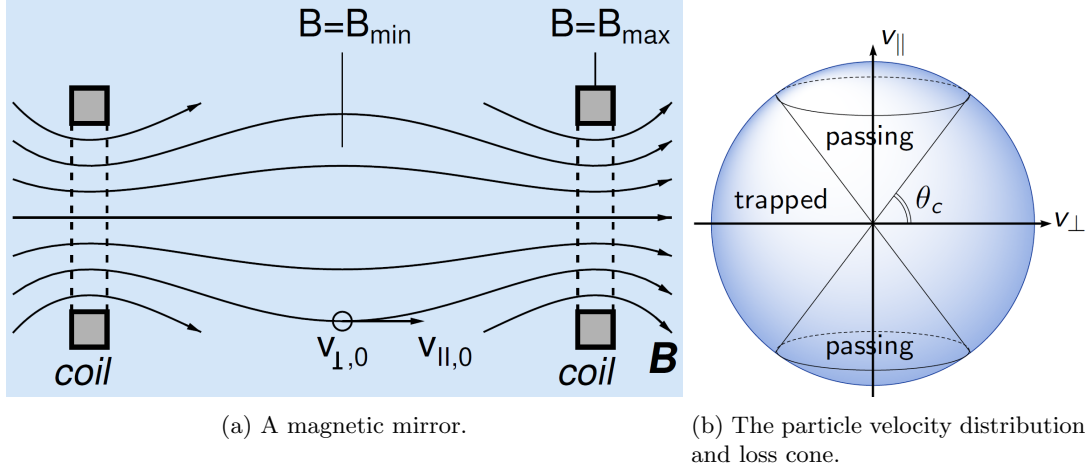


Figure 14: Curvature B drift.

Example: Trapped particle fraction (Exercises 2-2)

2. Trapped particle fraction

Derive that the trapping fraction equals $f = \sqrt{1 - \frac{B_{min}}{B_{max}}}$.

Solution: The trigonometric relations for the triangle with orthogonal sides $v_{\parallel}$, $v_{\perp}$ and hypotenuse $v = \sqrt{v_{\parallel}^2 + v_{\perp}^2}$ are

$$\tan(\theta_c) = \frac{v_{\parallel}}{v_{\perp}} = \sqrt{\frac{B_{max}}{B_{min}}} - 1, \quad \cos(\theta_c) = \frac{v_{\perp}}{v} = \sqrt{\frac{B_{min}}{B_{max}}}, \quad \sin(\theta_c) = \frac{v_{\parallel}}{v} = \sqrt{1 - \frac{B_{min}}{B_{max}}}.$$

Banana orbits

¹² This magnetic effect also works when looking at a cross-section of a Tokamak. Remember

$$B_t \propto \frac{1}{R} \rightarrow B_{max} = \frac{B_0 R_0}{R_0 - r} \text{ and } B_{min} = \frac{B_0 R_0}{R_0 + r} \quad (25)$$

thus on the outboard side the magnetic strength is lower. This enables a mirror effect which makes “banana orbits” (see figure 15a). Which has the trapping condition

$$E_{kin} < \mu B_{max} \quad (26)$$

¹²Based on the slide 6, 24-37. See also Freidberg chapter 14.4.2-14.4.3

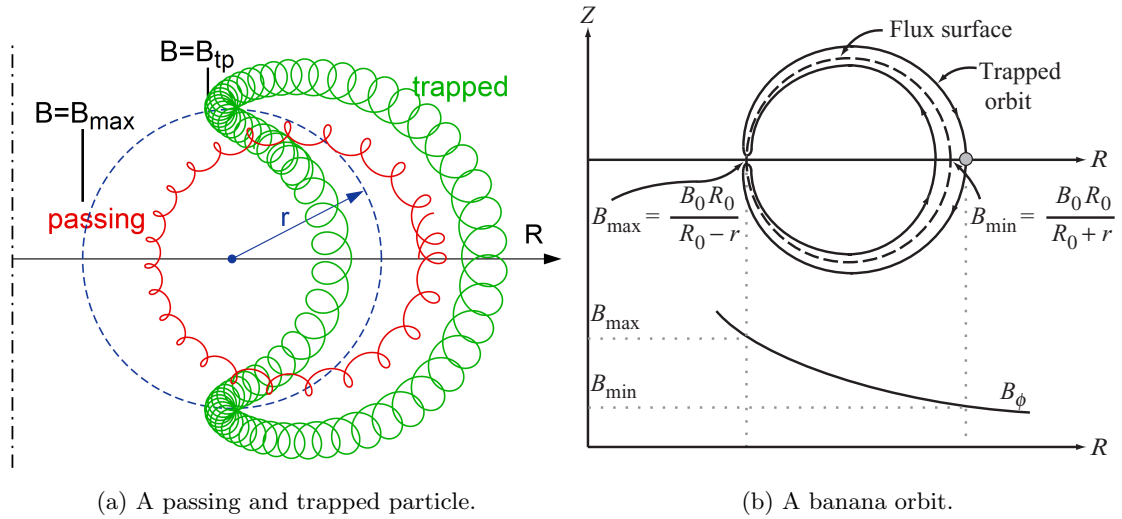


Figure 15: Diagrams of banana orbits.

Add example how to see if particle is trapped (past exam)

Example: Banana orbits (Exercises 2-3)

3. Banana orbits

- (a) Derive from $B \propto \frac{1}{R}$ that $\frac{B_{\max}}{B_{\min}} = \frac{R_0 + r}{R_0 - r}$.

Solution: The circle with center at $R = R_0$ and radius r has maximum field at its smallest major radius $R = R_0 - r$, so $B_{\max} = \frac{B_0}{R_0 - r}$. The minimal strength of the field is at the largest major radius $R = R_0 + r$, so $B_{\min} = \frac{B_0}{R_0 + r}$.

- (b) Derive the trapped particle fraction $f = \sqrt{\frac{2r}{R_0 + r}}$.

Solution: From exercise 2 one derives that

$$f = \sqrt{1 - \frac{B_{\min}}{B_{\max}}} = \sqrt{1 - \frac{R_0 - r}{R_0 + r}} = \sqrt{\frac{2r}{R_0 + r}}.$$

- (c) Derive the equation (*) for the banana width at the trapping boundary $\Delta r_{\max} = 2\sqrt{\frac{2r}{R_0}} \frac{B}{B_{\text{pol}}} \rho_c$ using $r \ll R_0$.

Solution: The step

$$\Delta r = 2 \frac{mv_{\parallel}}{qB_{\text{pol}}} = 2 \frac{v_{\parallel}}{v_{\perp}} \frac{B}{B_{\text{pol}}} \rho_c$$

follows from the expression for the gyroradius $\rho_c = mv_{\perp}/qB$.

At the trapping boundary we have

$$\frac{v_{\parallel}}{v_{\perp}} = \sqrt{\frac{B_{\max}}{B_{\min}} - 1} = \sqrt{\frac{R_0 + r}{R_0 - r} - 1} = \sqrt{\frac{2r}{R_0 - r}} \approx \sqrt{\frac{2r}{R_0}}.$$

Lecture 3: MHD introduction

Based on the slides. See also Freidberg chapter 10, 11.1-11.2, 11.7.1-11.7.6, 12.2.2, 11.5-11.6 and 13.7-13.8

Plasma distribution function

¹³ Distribution functions can be used to describe how particles behave. From which you can calculate properties, etc. For example

$$\boxed{dN = f(\vec{r}, \vec{v}, t) d^3r d^3v} \quad (27)$$

Here f is a particle distribution function, resulting in dN particles in the phase space $d^3r d^3v$ (space in both position and velocity of those relevant particles).

With this you can calculate a few macroscopic quantities:

- Particle density: $n(\vec{r}, t) = \int f d^3v$
- Mean velocity: $\langle \vec{v}(\vec{r}, t) \rangle = \vec{u}(\vec{r}, t) = \frac{\int \vec{v} f d^3v}{\int f d^3v}$
- Particle flux $\boxed{\Gamma(\vec{r}, t) = n\vec{u} = \int \vec{v} f d^3v}$
- Kinetic energy density $w = n \langle \frac{1}{2} m v^2 \rangle = -D \nabla n = \frac{1}{2} m \int v^2 d^3v = \frac{3}{2} p + \frac{1}{2} m u^2 n$ (where the first term is internal energy from pressure and the second term is flow energy).

The plasma can have a certain flux which results in a change over time

$$\boxed{\frac{\partial n_s}{\partial t} + \nabla \cdot \vec{\Gamma}_s = S_s} \quad (28)$$

where $\vec{\Gamma}$ is the flux, D is the diffusion coefficient, S are any sources and $_s$ indicated a specific species. The question can also be formulated for all species by making each term a sum weighted by the mass of the species. That results in the conservation of mass, while doing it with charge would result in the conservation of charge.

The path of these particles can be effected by external forces (such as the Lorentz force) or collisions. The Boltzmann equation this can take this into account

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla_r f + \frac{\vec{F}}{m} \cdot \nabla_v f = \left(\frac{\partial f}{\partial t} \right)_{\text{collision}} \quad (29)$$

in the case of a steady-state equilibrium one solution would be the (shifted-) Maxwell-Boltzmann distribution

$$f_s(\vec{v}) = n_s \left(\frac{m_s}{2\pi k_B T_s} \right)^{\frac{3}{2}} \exp \left(-\frac{m_s}{2k_B T_s} |\vec{v} - \vec{u}_s|^2 \right) \quad (30)$$

with three main parameters (see figure 16)

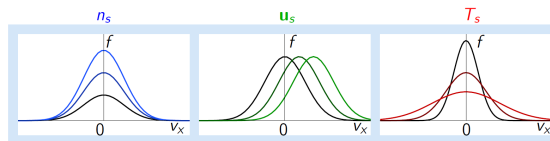


Figure 16: The 3 parameters of the Maxwell-Boltzmann distribution.

¹³Based on slide 6-37. See also Freidberg chapter 10. Also the course 3MB010 Physics of plasma radiation

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slide 12

You can also take the moments of the distribution

$$A = \int \alpha f d^3v \quad (31)$$

where α is the moment and A is the resulting quantity. The power of the velocity determines the moment number.

Table 1: Moments of the Maxwell-Boltzmann distribution.

Moment	A	α
0	particle density n	1
1	particle flux $\vec{\Gamma}$	$\vec{v}$
1	momentum $m\vec{u}$	$\vec{v}$
2	pressure $p = nT$	$\frac{m}{3}(\vec{v} - \vec{u})^2$
2	energy	$\frac{m}{2}\vec{v}^2$
3	heat flux $\vec{q}$	$\frac{m}{2}(\vec{v} - \vec{u})^3$

Furthermore if you take the moment of the Boltzmann equation you will get a time derivative

$$\frac{\partial A}{\partial t} = \int \alpha [\text{eq}] d^3v \quad (32)$$

where [eq] is the Boltzmann equation 29. This can be used to prove particle conservation, momentum conservation and energy conservation.

Ideal MHD

¹⁴ The Magneto-hydrodynamics (MHD) is a way to model plasmas via fluid dynamics. It is based on the following assumptions:

- $m_i \gg m_e$
- Neglect displacement current
- Neglect gyration: $t \gg \omega_{ce}^{-1}$ and $L \gg \rho$
- Assume quasi-neutrality: $t \gg \omega_p^{-1}$ and $L \gg \lambda_{Debye}$
- Neglect collisions but assume local thermodynamic equilibrium

In the end the ideal force balance MHD equation is

$$\rho \frac{D\vec{u}}{Dt} = \frac{1}{\mu_0} \vec{B} \cdot \nabla \vec{B} - \nabla \left(p + \frac{b^2}{2\mu_0} \right) \quad (33)$$

with the conservation laws

$$\begin{aligned} \frac{d}{dt} \int_V \rho d^3x &= 0 \text{ (mass)} \\ \frac{d}{dt} \int_A \vec{B} \cdot d^2\vec{x} &= 0 \text{ (magnetic flux)} \\ \frac{D}{Dt} \left(\frac{p}{\rho^{5/3}} \right) &= 0 \text{ (entropy)} \end{aligned} \quad (34)$$

where ρ is the density, $\vec{u}$ is the (average) velocity of the plasma, $\frac{D}{Dt} = \frac{\partial}{\partial t} + \vec{u} \cdot \nabla$

¹⁴Based on slide 38-48. See also Freidberg chapter 11.1-11.2.

With the complete equations

$$\begin{aligned}
 \text{mass :} \quad & \frac{d\rho}{dt} + \rho \nabla \cdot \mathbf{v} = 0; \\
 \text{momentum :} \quad & \rho \frac{d\mathbf{v}}{dt} = \mathbf{J} \times \mathbf{B} - \nabla p; \\
 \text{Ohm's law :} \quad & \mathbf{E} + \mathbf{v} \times \mathbf{B} = 0 \quad \text{ideal MHD,} \\
 & \mathbf{E} + \mathbf{v} \times \mathbf{B} = \eta_{\parallel} \mathbf{J} \quad \text{resistive MHD;} \\
 \text{energy :} \quad & \frac{d}{dt} \left(\frac{p}{\rho^{\gamma}} \right) = 0; \\
 \text{Maxwell :} \quad & \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\
 & \nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \\
 & \nabla \cdot \mathbf{B} = 0.
 \end{aligned}$$

Figure 17: Basic MDH model.

The MHD equations have several energy sources/sinks

- Energy exchange (between species/particles).
- Ohmic heating
- RF and NB heating
- Radiation losses
- Fusion power
- Ionization
- Viscous friction

Misschien
verwi-
jderen/ver-
vangen.

Maybe add slide 48

Maybe add slide 40, Freidberg 12.2.2

lecture 4: MDH Tokamak equilibrium

Based on the slides. See also Freidberg chapter 12.1-12.3.3, 12.4-12.5 and 13.7.3-13.7.4

15

On short time-scales the ideal MHD equilibrium is determined only by the plasma pressure and magnetic force

$$\nabla p = \vec{J} \times \vec{B}, \text{ with } \vec{J} = \frac{1}{\mu_0} \nabla \times \vec{B} \quad (35)$$

Something about Hopf's theorem, slide 54.

This implies that if the magnetic field

- is axisymmetric and
- purely toroidal

that there would be no confinement. Thus a twisted field is needed. There are three sources of rotational transform:

- Toroidal current (Tokamaks)
- Torsion of the magnetic axis (figure-8 Stellarator)
- Twisted cross section (complex Stellarators)

Maybe describe more

Figure from slide 58

Safety factor

¹⁶ The safety factor is

$$q = \frac{\text{Toroidal turns}}{\text{Poloidal turns}} = \frac{r B_t}{R B_p} \quad (36)$$

as you can see this increases with r . This causes a magnetic shear

$$s = \frac{r}{q} \frac{dq}{dr} \quad (37)$$

Hoort die grote R onderaan? Volgens lecture 6 wel?

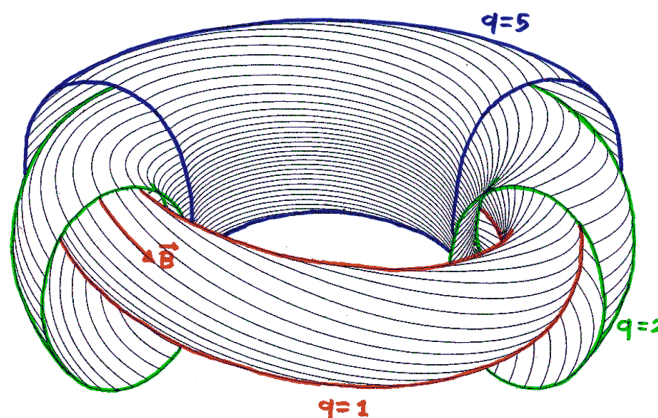


Figure 18: Different typical values of q .

¹⁵Based on slide 49-60. See also Freidberg chapter 11.5-11.7

¹⁶Based on slide 67-68. See also Freidberg chapter 11.5-11.7

Grad-Shafranov

¹⁷ The Grad-Shafranov equation is the equation for a MHD equilibrium in a axis-symmetric plasma

$$\nabla^* \psi = -R^2 \frac{dp}{d\psi} - F \frac{dF}{d\psi} \quad (38)$$

where ∇^* is a special Laplacian-like operator, p is the pressure profile, ψ is a specific flux function of the magnetic field, F is an arbitrary function related to the toroidal magnetic field (you don't need to remember this).

For a large aspect ratio approximation ($R \gg a$) the plasma can be approximated as a cylinder which has a certain solution for the Grad-Shafranov equation. It results in a horizontal force balance which results in the Shafranov shift where the center of the plasma shifts outwards due to tire tube force or Hoop force (discussed later in lecture 6 Tokamaks).

Exercise 9 maybe?

¹⁷Based on slide 60-66, 68-74. See also Freidberg chapter 11.5-11.7

lecture 5: MHD instabilities

Based on the slides. See also Freidberg chapter 12.1-12.3.3, 12.4-12.5 and 13.7.3-13.7.4

MAKE THIS, THIS IS CURRENTLY NOT FINISHED

Balloning mode: <https://gemini.google.com/app/3c16c466259832cd>

Sawtooth: <https://gemini.google.com/app/6b637d3f176f877a> Internal kink:

<https://gemini.google.com/app/d57b4b02acfaf9e2>

Linear waves

Hier slide 71 tot en met

Z-pinch kink instability

When considering/approximating the plasma confinement as a cylinder there can be several instabilities:

- Kink instability: if the plasma is bent slightly the magnetic field will be higher on the side with the bend and lower on the side without the bend thus increasing the bend.
- Sausage instabilities: if the plasma is restricted that will result in a higher magnetic field resulting in more constriction.

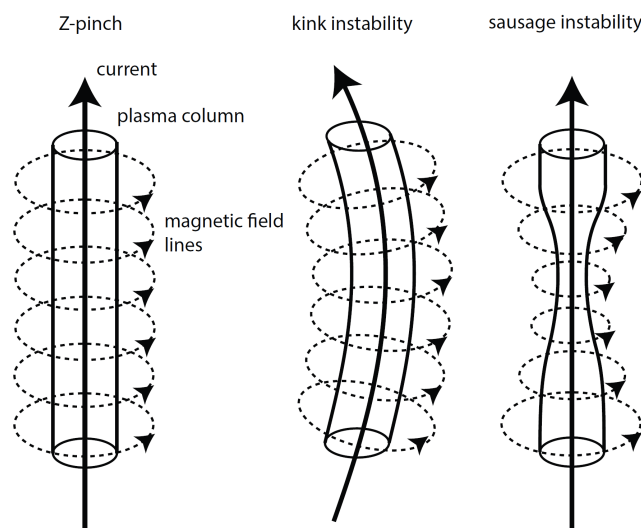


Figure 19: The instabilities with the z-pinch.

Energy functional

afmaken

The energy functional is minimized for $q = \frac{m}{n}$, this means instabilities are localized at resonant magnetic surfaces.

Current driven instabilities

- External kink modes for the part of the torus with $q > \frac{m}{n}$
- Tearing modes for parts of the torus with $q < \frac{m}{n}$ due to magnetic reconnection forming magnetic islands.
- TAE modes
- Internal kink modes for part of the torus with $q < 1$

Pressure driven instabilities

For a disturbance in the magnetic field we find:

$$\vec{B}_{dis} = \nabla \times (\zeta \times \vec{B}_0) \quad (39)$$

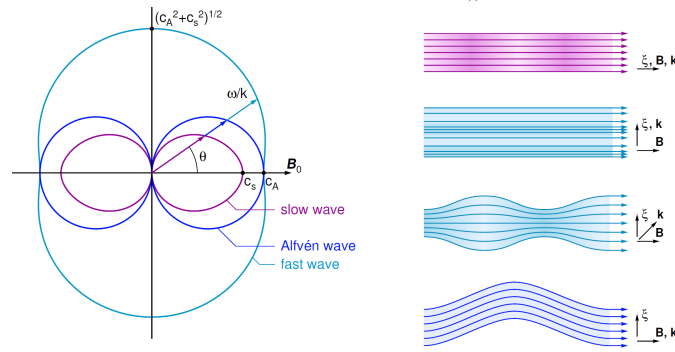


Figure 20: A Friedrich diagram with different waves.

MDH waves, also consider implimenting exam 2024 apr problem 21



Types of current driven instabilities

Localization near a resonant surface $r = r_{m/n}$ where $q \approx \frac{m}{n}$, restricts the possible current-driven instabilities:

- **External kink modes** affect all plasma $r > r_{m/n}$ extending to the plasma edge. The $m = 2$ external kink mode is unstable if $q(a) < 2$!
- **Tearing modes:** resistivity $\eta > 0$ enables modes localized at $r = r_{m/n}$. Reconnection forms magnetic islands.
- **TAE modes** (toroidicity induced Alfvén eigenmodes) are Alfvén waves driven unstable by a population of resonant energetic particles.
- **Internal kink mode:** $m = n = 1$, affects all plasma within $q < 1$.

108/146

Hugo de Blank, MHD

Figure 21: Types and causes of current driven instabilities.

Two islands (m_1, n_1) and (m_2, n_2) cause a **secondary island** with summed mode numbers, so its q -value is between the primary islands,

$$\frac{m_1}{n_1} < \frac{m_1 + m_2}{n_1 + n_2} < \frac{m_2}{n_2}$$

For a monotonic $q(r)$ profile we have $r_1 < r_{12} < r_2$.

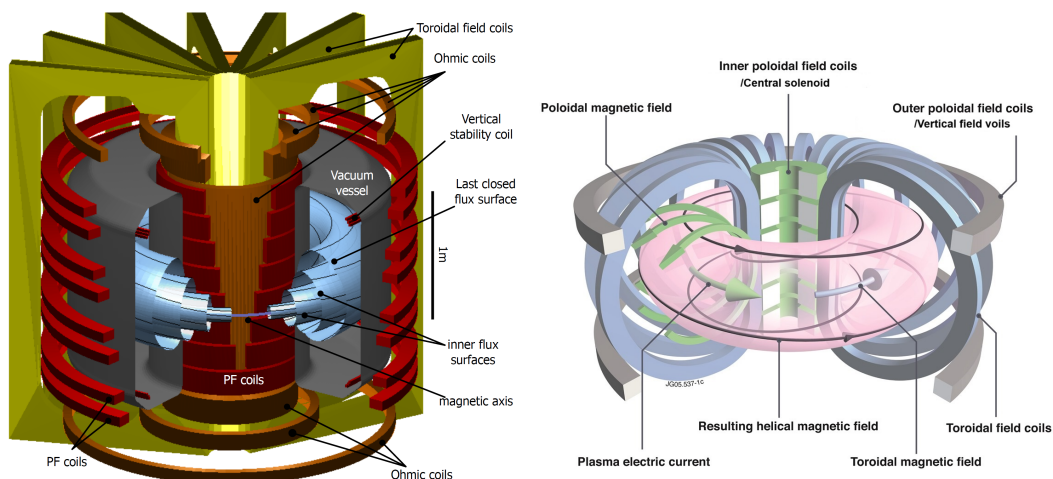
Figure 22: Secondary islands.

Lecture 6: Tokamaks

Based on the slides and parts of 4SC010 Control and Operation of Tokamaks.

A Tokamak has many magnetic coils (see figure 23). They are:

- Toroidal field coils: placed on the cross section of the torus, they serve to make the confining toroidal (direction of plasma) fields and are the most basic part of the confinement.
- Central solenoid/Ohmic coils: placed in the center, is there to induce a plasma current. Similar to the poloidal field coils.
- Poloidal/vertical field coils: places parallel to the donut, is there to make the poloidal field which is needed for the helical B-field. Also used for vertical stability and thus sometimes coils are placed inside the vessel for better reactivity, these are called plasma control coils.
- Shaping/divertor coils: extra coils parallel to the donut (around the divertor) used to better shape the plasma.
- Error field coils: placed as squares on the vessel to correct for misalignment's or manufacturing errors/tolerances.



(a) A overview of the varies coils in a Tokamak. (b) A overview of the basic fields (and coils) in a Tokamak.

Figure 23: The magnets and fields in a Tokamak.

When thinking of these problems and the direction of currents/magnetic fields keep in mind the right-hand-rule (see figure 24) which works both for determining the magnetic field or the induced current.

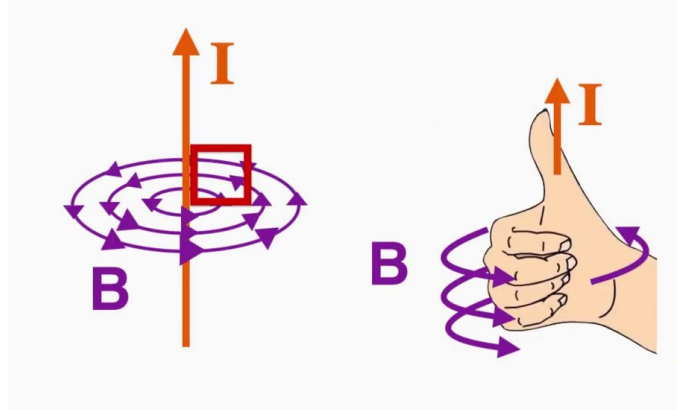


Figure 24: Right hand rule for magnetic field generation/induction.

Toroidal field

The toroidal field is responsible for the confinement since it reduces the Lamor-radius/step-size and thus transport

$$\tau_E \propto n^{-1} T^{-0.5} B^2 \quad (40)$$

this often is the limiting factor by superconducting properties (around max 13 or 6 Tesla).

Poloidal field

As discussed in the previous chapter a poloidal field is needed to counteract the drifts with a helical structure. This is described by q the safety factor (see equation 36) it can also be described in terms of the poloidal flux

$$q = \frac{\text{Toroidal turns}}{\text{Poloidal turns}} = \frac{d\Phi}{d\Psi} \approx \frac{r B_\phi}{R B_\psi} \quad (41)$$

where Φ is the toroidal flux and Ψ the poloidal flux.

Example: Calculations from q (exam April 2019 problem 3c)

Consider a tokamak with minor radius $r = a = 1.0 \text{ m}$ and major radius $R_0 = 5.0 \text{ m}$. The plasma cross section is circular. At the magnetic axis the electron density is $n_e(0) = 10^{20} \text{ m}^{-3}$, the temperature $k_B T(0) = 10 \text{ keV}$, and the magnetic field is $B_0 = 5.0 \text{ T}$.

You may assume the plasma has a large aspect ratio ($a \ll R$) for convenience, so that the toroidal field is $B_\phi(R) \approx B_0$. Give numerical answers in SI units.

[6 pt] The q -profile is given by $q(r) = 0.9 + 1.3(r/a)^2$.

- (i) Express the poloidal field B_θ as a function of r .

Solution: $B_\theta(r) = B_\phi r / qR \approx B_0 r / qR_0 = 1.0 \text{ T} \times ra / (0.9a^2 + 1.3r^2)$

- (ii) Compute the total toroidal plasma current I .

Solution: $I = 2\pi a B_\theta(a) / \mu_0 = 2\pi(1 \text{ m})(0.45 \text{ T}) / \mu_0 = 2.3 \text{ MA}$

- (iii) Compute the current density on the magnetic axis.

Solution: Near the magnetic axis you can assume $r \ll R$, so 'cylindric geometry' like in a screw-pinch. At radius r , the enclosed current is $I(r) \approx \pi r^2 J_t(0)$. This is related to the poloidal field by $I(r) = \frac{2\pi}{\mu_0} r B_\theta(r)$. Using $B_\theta(r) \approx B_0 r / q(0) R_0$ this becomes $I = \frac{2\pi}{\mu_0} B_0 r^2 / q(0) R_0$, so $J_\phi = 2B_\phi / \mu_0 q(0) R_0 = 1.7 \text{ MA/m}^2$.

Improve

Slide 19 over shafranov shift

The poloidal field also is responsible for the plasma current. It induces a electric field/voltage which the causes the current

$$V_{loop} = -\frac{d\Psi}{dt} = RI_p \quad \text{with} \quad R = \frac{2R_0}{a^2} \eta \quad (42)$$

where η is the Spitzer resistivity $\eta \propto T^{-\frac{3}{2}}$. The current usually is very high (several MA) while the voltage is very low (singular Volts).

However to keep this current positive the magnetic flux (and thus magnetic field) need to continuously increase. After a while the coils which generate this field can't handle more current and are forced to shut down. The maximum amount of flux available is

$$\Delta\Psi = 2BA_{core} = L_p I_p + \int (V_{loop} - V_{selfInductance}) dt \quad (43)$$

where L_p is the inductance of the plasma. The first term is related to the start up of the reactor while the second term (of which the self-inductance can be neglected) represents the so called flat-top time $\Delta t_{flat-top}$ (during which the reactor actually operates).

Example of calculating flat top time

Operational limits

There are limits to the operational space of Tokamaks. Those are:

Current/q limit

If your safety factor gets to low magnetic islands will develop. If it is higher these will be small and not dangerous. Thus

$$q_a > 2 \quad (44)$$

since B_p is linear in the plasma current this is a current limit.

- Limits current
- Countered with larger a

Maybe add slide 23

Density/Greenwald limit

Radiation in plasma is a two body process and thus scales as n^2 . If the density thus gets to high the radiation power will become to high. This (initially) results in a cooler edge and thus a potentially a disruption. The empirical Greenwald/density limit is

$$n < \frac{I_p}{\pi a^2} \quad (45)$$

- Limits density.
- Countered with smaller a or larger I_p

Beta-limit

The plasma needs to stay confined so the kinetic pressure needs to stay below the magnetic pressure. In reality it should stay well below. We thus have the beta/Troyon limit

$$\beta < 3.5\% \cdot \frac{I_p}{aB_T} \quad (46)$$

- Limits pressure/temperature.
- Countered with larger I_p or smaller a .

Example: Non-variable scaling law parameters (exam April 2024 problem 8 and April 2018 problem 1.b.4)

8 For an efficient reactor a reduction of the transport or, equivalently, an increase in the energy confinement time is required. An empirical scaling law states:

$$\tau_E = 0.047 f_H M^{0.5} I^{0.85} B^{0.2} R^{1.2} a^{0.3} \kappa^{0.5} P^{-0.5} n_e^{0.1}$$

where f_H is the H-mode improvement factor, M the average amu mass of the plasma, I is the plasma current, B the toroidal magnetic field, R the major radius, a the minor radius, κ the elongation, P the heating power, n_e the electron density.

These values cannot be varied independently due to some MHD stability constraints, or cannot take all values due to engineering constraints. Mention 4 constraints, with 1 or 2 sentences each, explaining the physics/engineering background of the constraint.

Model answer

- q-limit, $q < 2$ limit on maximum I/B value,
- superconduction limit: Maximum B value at the coil,
- Greenwald limit: limit on maximum density, related to edge radiation,
- beta limit: limits plasma pressure/magnetic pressure, due to instabilities,
- limit on elongation: plasma unstable

Improve

Radial force balance

There are several outward forces on the plasma

- The Hoop force ($\propto B^2 \Delta S \propto I_p^2 \Delta S$ see figure 25a) is an outward force caused by the fact that the toroidal (parallel) magnetic field is stronger on the inboard side due to it having less surface area and that the strength of the field scales stronger than the surface area.
- The tire tube force ($\propto p \Delta S$ see figure 25b) is a similar outward force caused by the surface area on the inside being smaller while the pressure is the same.
- The 1/R force ($\propto B_t \frac{1}{R}$ see figure 25c) is also an outward force caused by the $\frac{1}{R}$ dependence of the toroidal field

Moet
dt niet
poloidal
field zijn?

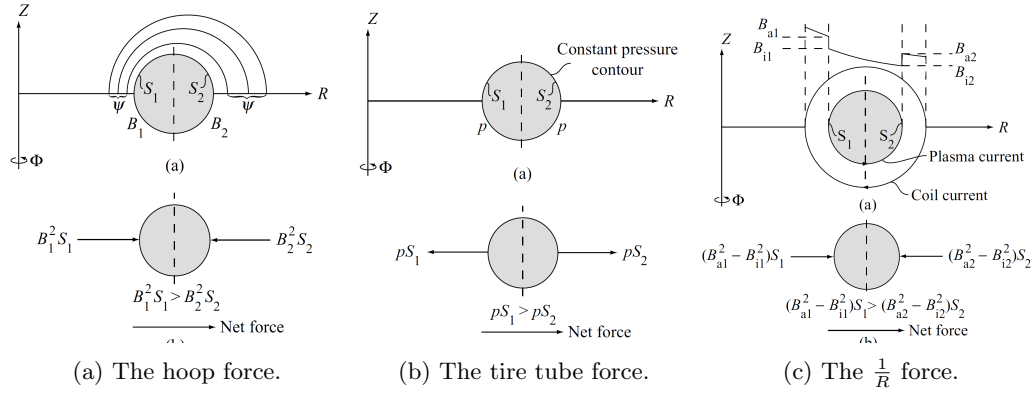


Figure 25: Different forces in a Tokamak.

Elongation

One can elongate the plasma (see figure 26). Elongation the the vertical direction increases the safety factor q proportionally

$$q_{elong} = q_{circ} \frac{b}{a} \quad (47)$$

Plasma shaping/control

Elongation, while beneficial for the safety factor, causes vertical stability problems. This causes the Tokamak to require vertical stabilizing coils.

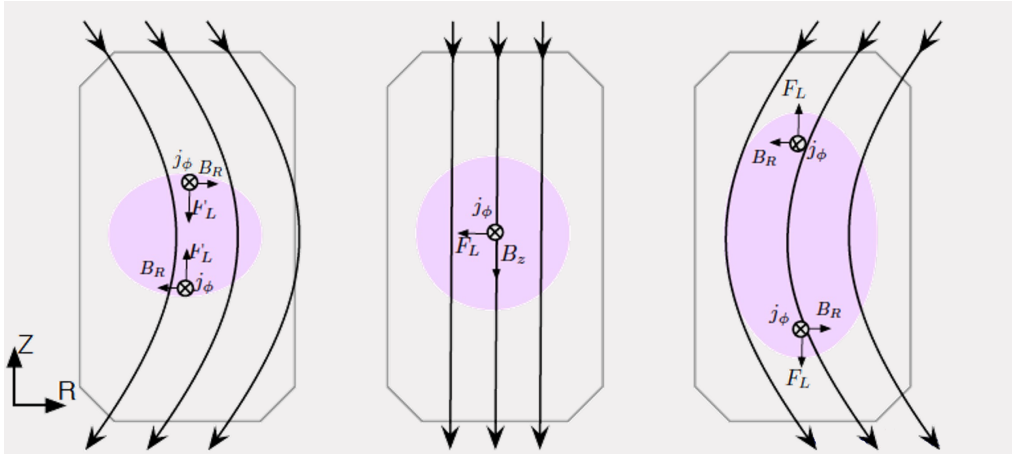


Figure 26: Different plasma elongations and their stability.

There are many ways to use the vertical coils. But the basics can be seen in figure 27, note the direction of the currents and forces.

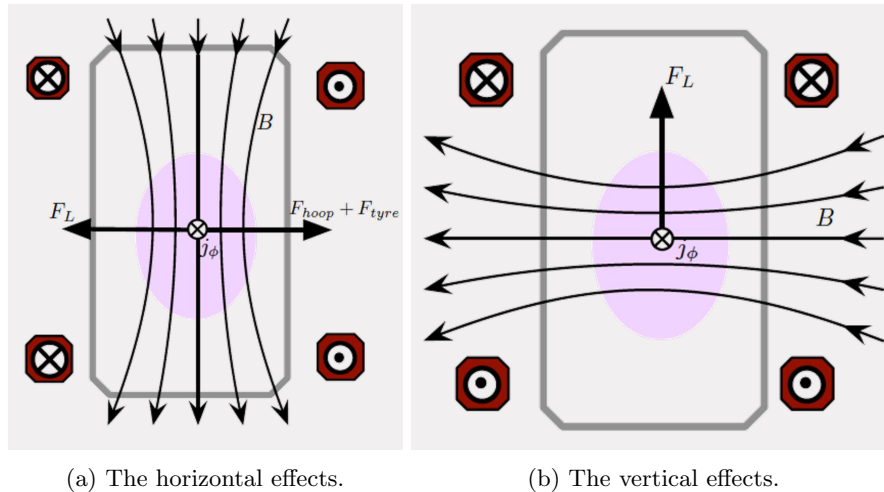


Figure 27: Different effects of plasma control with vertical coils.

Divertor

There needs to be an interface of where the plasma ends. This can be a limiter directly in the plasma. But to prevent impurities and remove population a divertor is used (see figure 28). This is created by placing a extra coil under the torus which causes an x-point.

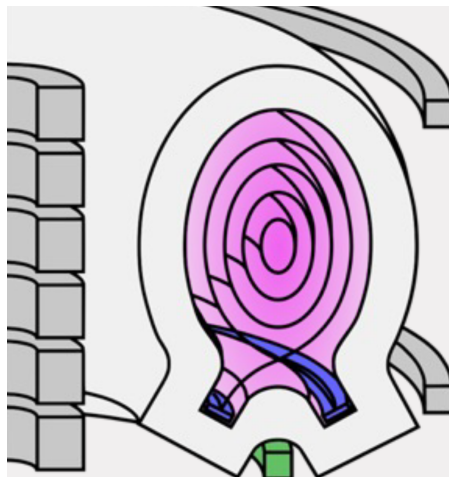


Figure 28: The concept of a divertor.

Plasma/Tokamak stages

The required voltage for a plasma breakdown depends on the pressure and distance of the loop pd (see the Paschen curve in figure 29). The distance here is often many times (up to 50 or more) the circumference of the Tokamak due to the helical nature of the magnetic field, thus increasing the required breakdown voltage but at the same time also multiplying the effective loop voltage. This voltage must be induced by the central solenoid.

Voorbeeld
berekening
toevoegen
zie slide 43

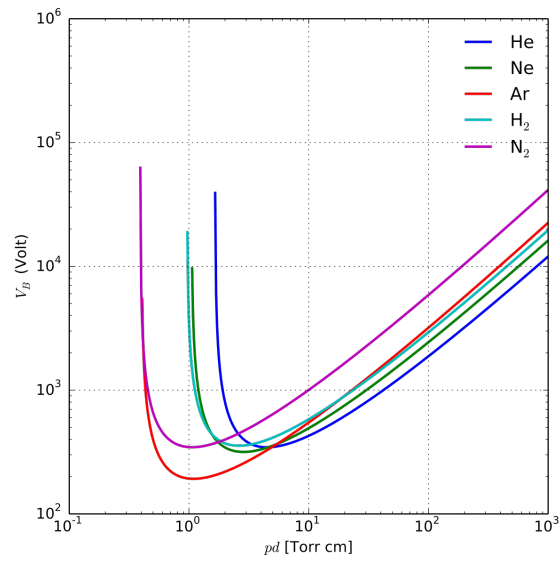


Figure 29: The Paschen curve.

At the conditions (and temperatures) of a Tokamak nearly all the particles will ionize and thus the ionization fraction $\frac{n_i n_e}{n_0}$ is very high.

Next you have the burn through where the temperature builds up to crease burn conditions.

After this you have the current ramp to build up the magnetic field.

Finally you have the flat-top pulse/current which has a limited length due to constantly increasing current needed in the central solenoid to maintain the induced plasma current. You can calculate the required time with equation 43 (and Ohms law).

Ambipolarity definition, ions and electrons dirft at the same rate.

Add how you can calculate V_{loop} and R

Lecture 7: Design

All covered in report, likely not needed for exam.

Lecture 8: Stellarators

Based on the slides. And Fusion Physics IAEA 8.2 & 8.4

A reactor needs rotation, that can be provided by a plasma current as in a Tokamak, or as they do in a Stellarator with rotating poloidal cross-sections and rotating magnetic axis.

There can still be magnetic islands in such reactors if the safety factor is rational $q = \frac{m}{n}$ due to resonance. The island size is proportional to

$$w \propto \frac{1}{ml} \quad (48)$$

where l is the rotation. Thus there are two strategies:

- Don't avoid resonances but use high shear (change in q) to power through all regions which can form islands thus minimizing their size (high l).
- Avoid low order rationals thus also minimizing the size of the islands (high m).

Iets over Pfirsch-Schlüter current (slide 23 and 51)

The nice thing about Stellarators is that they have different operational limits

- No Greenwald density limit.
- No current limit (since no current).
- Less strict pressure/beta limit.

Currents in stellarators

Equilibrium: $\vec{j} \times \vec{B} = \nabla p \rightarrow \vec{j}_\perp = \frac{\vec{B} \times \nabla p}{B^2}$

Quasineutrality: $\nabla \cdot \vec{j} = 0 \rightarrow \nabla \cdot \vec{j}_\perp = -\nabla \cdot \vec{j}_\parallel$

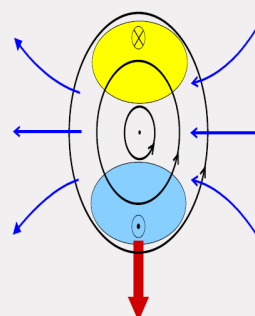
$$\nabla \cdot \vec{j}_\parallel = -\frac{2p'}{B^3} (\nabla r \times \vec{B}) \cdot \nabla B = -\frac{2p' |\nabla r|}{B} \kappa_g$$

Simple toroidal configuration $B = \frac{B_0}{1 + \frac{r}{R} \cos \vartheta}$

$j_\parallel = -\frac{2p'}{\iota B_0} \cos \vartheta$

Pfirsch-Schlüter current

Diamagnetic current



TU/e

51

Figure 30: Pfirsch-Schluter current.

Lecture 9: Classical transport

¹⁸ There are 3 types of transport in fusion

- Classical transport which is mostly based on fluid-dynamics/collisions and much lower than expected.
- Neo-classical transport which is based on collisions and drifts and much higher but still much lower than empirical, especially for electrons and Tokamaks.
- Turbulent/anomalous transport which is based on turbulence (thus very hard to predict) and which is on the same level as empirically seen

In (classical) transport one the most important equations is Fourier's law

$$\vec{q} = -n\chi\nabla T \quad (49)$$

where $\vec{q}$ is the heat-flux and χ is the (heat) conductivity. The heat conductivity directly relates to the confinement time $\tau_E = \frac{3a^2}{4\chi}$. We thus want it to be as low as possible.

For a uniform power absorption the temperature takes on a inverse parabola (with zero at the edge), while for a point/peaked source it will be a piece-wise temperature, constant and then linearly decreasing (see figure 31).

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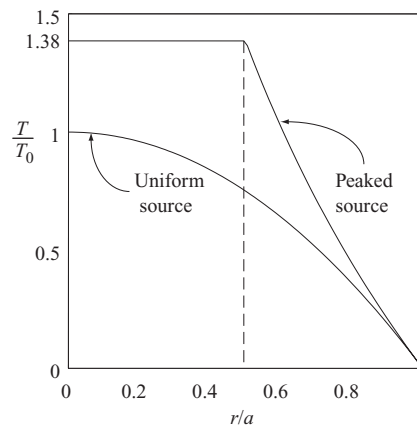


Figure 31: The temperature profile for different sources.

¹⁸Based on the slides. See also Freidberg chapter 14.1-14.3 and lecture notes

Example: Central temperature prediction with given gradient (exam April 2018 problem 2a)

[3 pt] To predict density and temperature profiles in fusion plasmas it is crucial to understand the transport of heat and particles.

Assuming ohmic heating only, derive how the central temperature T_0 scales with the heat diffusivity χ . You can assume that the temperature gradient can be approximated as follows: $\nabla T \approx T_0/a$ with a being the minor radius.

Solution: With heat flux q , density n , diffusivity χ , torus surface A , major radius R , minor radius a , current I , resistance $\mathcal{R}$, Spitzer resistivity $\eta = CT^{-3/2}$ with C a constant, length of torus l , poloidal cross section A_{pol}

$$\begin{aligned} P_{loss} &= qA = n\chi \nabla T A = n\chi \frac{T_0}{a} 4\pi^2 Ra \\ P_{\Omega heat} &= I^2 \mathcal{R} = I^2 \eta \frac{l}{A_{pol}} = I^2 \eta \frac{2\pi R}{\pi a^2} \\ P_{loss} &= P_{\Omega heat} \\ n\chi \frac{T_0}{a} 4\pi^2 Ra &= I^2 \eta \frac{2\pi R}{\pi a^2} \end{aligned}$$

Giving

$$T_0 = \left(\frac{CI^2}{2n\pi^2 a^2 \chi} \right)^{2/5}$$

And thus

$$T_0 \propto \chi^{-2/5}$$

improve

Example: Central temperature prediction (exam April 2019 problem 2a)

[6 pt] Consider a large aspect ratio, circular cross section tokamak, with major radius $R_0 = 5m$ and minor radius $a = 1m$. During steady state flat-top operation, power balance is dominated by thermal conduction losses and the external heating power $P_h = 50MW$.

Assume the thermal diffusivity $\chi = 1m^2/s$, number density $n = 10^{20}m^{-3}$ and the profile of the heating power S_h are uniform in space. Derive an expression for the central $k_b T_e(0) = k_b T(0) \equiv k_b T(0)$ as a function of χ , n , major radius R_0 , minor radius a and the total heating power P_h .

Solution: $\nabla \cdot q = S_h$; $q = -n\chi \frac{\partial k_b T}{\partial r}$

$$\begin{aligned} \frac{1}{r} \frac{d}{dr} r q &= -\frac{1}{r} \frac{d}{dr} \left(r n \chi \frac{\partial k_b T}{\partial r} \right) = S_h \\ -r n \chi \frac{\partial k_b T}{\partial r} &= \frac{1}{2} S_h r^2 + C_1; \frac{\partial k_b T(0)}{\partial r} = 0 \rightarrow C_1 = 0 \end{aligned}$$

$$\begin{aligned} \frac{\partial k_b T}{\partial r} &= -\frac{1}{2} \frac{S_h}{n\chi} r \\ k_b T(r) &= -\frac{S_h}{4n\chi} r^2 + C_2; k_b T(a) = 0 \rightarrow C_2 = \frac{S_h}{4n\chi} a^2 \end{aligned}$$

$$P_h = S_h \times Vol = 2\pi^2 a^2 R_0 S_h$$

$$\text{Result: } k_b T(r) = \frac{P_h}{8\pi^2 R_0 n \chi} \left(1 - \left(\frac{r}{a} \right)^2 \right)$$

$$\text{Therefore } k_b T(0) = \frac{P_h}{8\pi^2 R_0 n \chi}$$

$$\text{and } k_b T(0)[eV] = \frac{P_h}{8\pi^2 R_0 n \chi e} = 7.8keV$$

improve, consider replacing with apr 2024 problem 20 which is a bit simpler. Also consider using my solution.

Particle transport (and thus convection) is the other large part of transport. That can be predicted using Fick's law for particle flux

$$\vec{\Gamma} = -D\nabla n \quad (50)$$

where D is the diffusion coefficient.

Random walk model

Transport due to collisions can be modeled using a random walk

$$D = \frac{\langle (\Delta x)^2 \rangle}{\Delta t} = \frac{\text{stepsize}^2}{\text{collision time}} \quad \text{and} \quad \chi = \frac{\langle (\Delta x)^2 \rangle}{\Delta t} \quad (51)$$

which in the direction of B can be take as the mean free path.

This step-size differs in what direction of B we are considering:

- Transport along B : the stepsize is the mean-free path (very long).
- Transport across B : the stepsize is the lamor radius.

For the time difference one takes the collision time but

- Particle transport is dominated by unlike particles (so ions and electrons) colliding (with no center of mass change) and thus has $\Delta t = \tau_{ei}$.
- Heat transport works with like particles and thus has $\Delta t = \tau_{ii}$ or τ_{ee} . Which means that ions dominate due to their larger Larmor radius/banana orbit width (which is slightly compensated by the longer collision time). $\chi_e \approx D$ and $\chi_i \approx \sqrt{\frac{m_i}{m_e}} D$

Of is dit τ_{ie} ?

Example: Perturbation in magnetic field (exam April 2024 problem 11)

Sort the approximate size of the collisionalities.

$$\tau_{ie} > \tau_{ii} > \tau_{ee} \approx \tau_{ei}$$

Misschien iets toevoegen over proportionaliteit D (zie examen 2018 april probleem 2.B.4)

Toevoegen misschien plaatje slide 23

Experiments

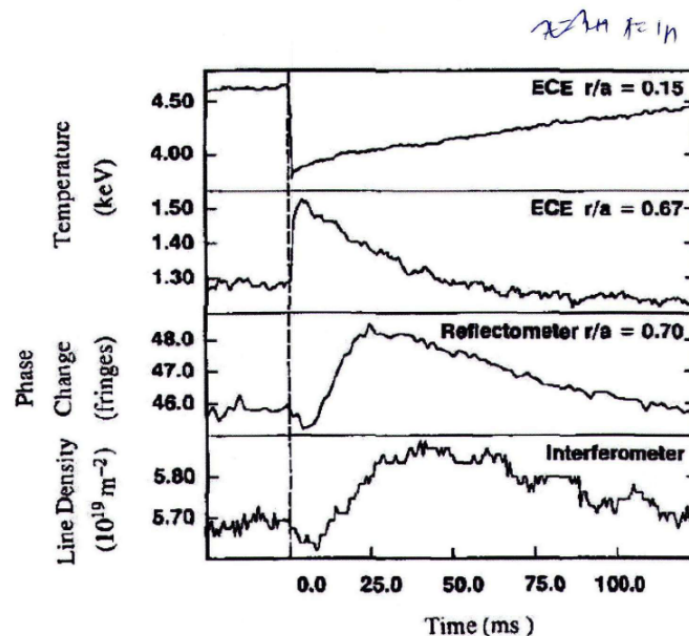
There are two major ways to measure the heat diffusivity

- Power balance where you keep track of power input, wait until in equilibrium and then measure the (core) temperature. Using Fouriers law you can then simply calculate the global diffusivity.
- Heat pulse propagation where heat is periodically (and locally) injected and the temperature is measured at different regions. Then by looking at the phase-lag and frequency at each location you can determine the local heat diffusivity.

Example: Heat pulse propagation during sawtooth at JET (exam April 2024 problem 19a & 19b)

The heat diffusivity χ can be measured in perturbative transport experiments, like modulated Electron Cyclotron heating. However, the sawtooth crash also produces a heat and density pulse from which the heat transport coefficient can be deduced.

Here you see an example of the time traces of the electron temperature and density for a sawtooth crash at JET.



With note that $a \approx 1$ meter

19a Explain the reason for the general difference in particle and heat diffusion.

Model answer

Particle transport lower due to ambipolarity constraint (1P) and the fact that like particle collisions do not lead to particle transport (1P). This is indicated since the density perturbation is much slower in this case: it takes a longer time for the density pulse to arrive a position at the outer region (1P).

Improve and fix line

Estimate the particle diffusion coefficient for this specific case.

Approximate using $D = \frac{\Delta r^2}{\Delta t}$

From graph:

- $\Delta t \approx 30$ ms
- Visible at 0.7 and inverse at 0.15 (indicating that the particles went there). So $\Delta r \approx (0.7 - 0.15) \cdot 1 = 0.55$ meter

This results in $D \approx \frac{0.55^2}{0.030} \approx 10$ m²/s.

Comparison with reality

Pagina 30 toevoegen tenzij dit ook besproken word bij lecutre 11.

Lecture 10: Neoclassical transport

19

One of the assumptions behind classical transport is a negligible mean-free path. This however is not completely the case thus with neoclassical transport these drifts are also taken into account. Trapped versus passing particles have effect so need to be taken into account separately

Passing particles

For diffusion it is found that

$$D_n^{NC} = 4q^2 D_n^{CL} \quad (52)$$

so it increases by a the safety factor squared at least compared to classical. Similarly it is the case for the neoclassical heat diffusion.

Trapped particles

Trapped particles cause way more transport because of their banana orbits, which is an disadvantage. While passing particles are much smaller due to their drift velocity. Specifically their step size is the entire banana width is a factor of $\frac{q}{\sqrt{\epsilon}}$ larger than passing

$$D_{trap}^{NC} = \epsilon^{-\frac{3}{2}} D_n^{NC} \quad (53)$$

where $\epsilon = \frac{r}{R}$ is the inverse aspect ratio. This is approximately a factor 100 larger than the passing diffusion. The heat diffusion scales the similarly.

These equations assume a low rate of collisions compared to the transit frequency of the trapped particles (banana orbit duration), this is the case for most Tokamaks but not always. There are three regimes (see figure 32a):

- Banana regime where the particle gets the chance to complete orbits and thus the the step size can be the full banana radius.
- Plateau regime where the transport remains approximately constant with increased collisionality. “more collisions means more steps” is evened out by the trapped particles not being able to complete their orbits anymore (and thus their step size decreasing).
- Pfirsch-Schlüter regime where so many collisions happen the banana orbit becomes irrelevant, here it becomes the same as passing particles.

The plateau disappears if the inverse aspect ratio is relatively small $1 \ll \epsilon$.

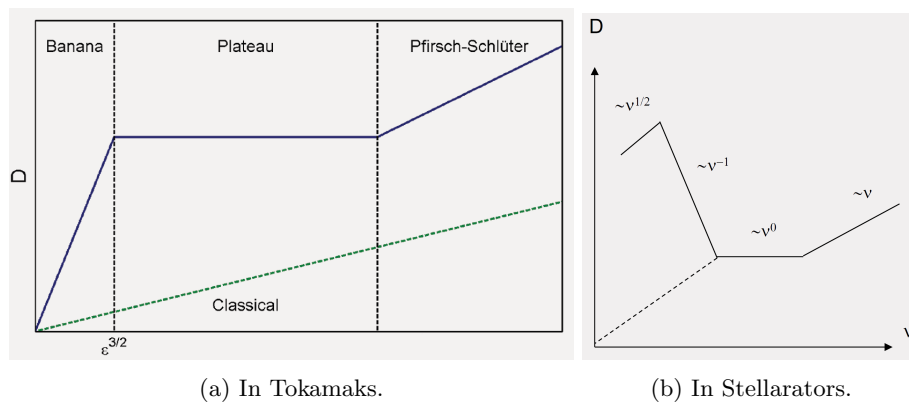


Figure 32: Neo classical transport versus collisionality.

¹⁹Based on the slides. See also Freidberg chapter 14.4 and lecture notes

Neoclassical transport in Stellarators

For Stellarators (see figure 32) the Banana regime is replaced by two sub-regimes (in reverse order):

- The ν^{-1} regime is caused because the banana orbits are not closed in a stellarator (and slowly drift to the outside).
- The $\nu^{\frac{1}{2}}$ regime is caused by trapped particles becoming “free”/passing due to drift caused by the electrical field.

A higher electric field increases the size of the $\nu^{\frac{1}{2}}$ regime and thus lowers the diffusion.

Bootstrap current

The bootstrap current is the most important prediction of neoclassical transport. The center is hotter and denser than the edge. Thus there are more particles in banana orbit with a higher velocity. This differential causes a net current. To understand consider two orbits of which their respective inner and outer part overlap, they each will produce a current in their direction of travel but the inner one has more and faster electrons (see figure 33).

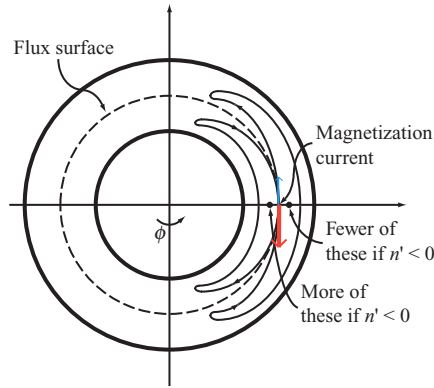


Figure 33: The bootstrap current (top view of Tokamak). Keep in mind that electrons flow opposite of the current.

Ware pinch

The Ware pinch (see figure 34) is another important prediction of neoclassical transport. The E-field that drives the current increases the (parallel) velocity on one-half of the banana orbit and decreases it on the other half. Thus it shifts the reflection points. it also causes an inward drift (for both electrons and ions).

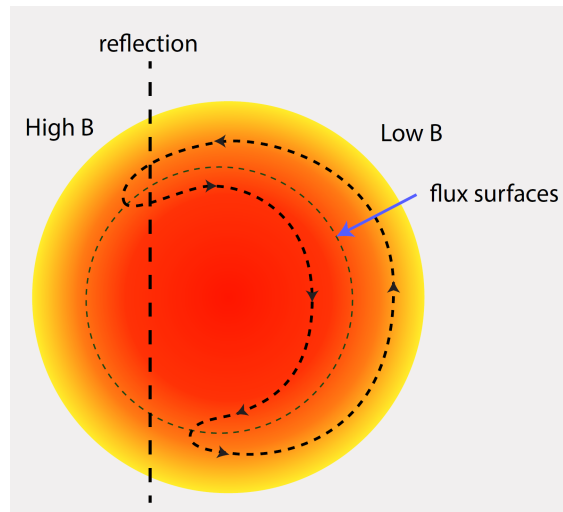


Figure 34: The Ware pinch.

Lecture 11: Turbulence

Still when comparing neoclassical to reality there seems to be a discrepancy of a factor 300. This difference is attributed to anomalous transport which later turned out to be turbulence.

Maybe add slide 30 from lecture 9 here

Turbulence is caused by non-linear chaos which grows. This turbulence disappears from large to small, this is called the Kolmogorov energy cascade (see figure 35) which says the amount of energy scales as $E \propto k^{-\frac{5}{3}}$.

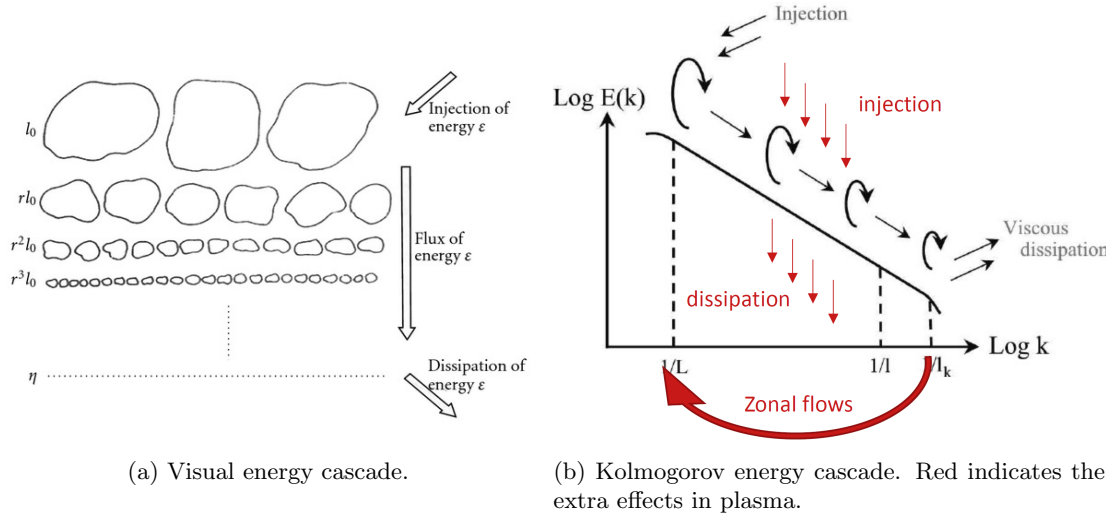


Figure 35: The energy cascade.

In a fusion plasma there is injection at different scales (due to different instabilities), there are also damped modes at many scales causing dissipation and finally interaction between different scales is possible due to zonal flows, which enable inverse cascades to larger scales.

But again a good way to investigate transport is with the step-size and time-step. For turbulence:

- The step-size is approximately the size of the eddies.
- The time-step is the correlation time of the turbulent eddies

Maybe add apr 2024 16a here.

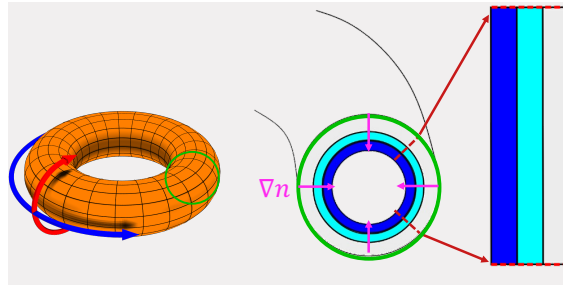


Figure 36: Graph to investigate the disturbances. A slice on the outboard side is taken.

One easy way to visualize/investigate disturbances is by taking a cross-section of a reactor and then “unrolling” (a slice of) the reactor cross-section (see figure 36). The x-axis is then the angular coordinate, the y-axis is the radius (with increasing y is more towards the center).

Drift waves

If this density gradient is perturbed (sinusoidally in figure 37), because electrons are more mobile and thus equalize faster this results in a charge difference and an electric field from extrema. This then causes a $E \times B$ force between extrema, which propagates the perturbation. This is called a drift wave. This has a *neutral stability* since it doesn't grow nor shrink.

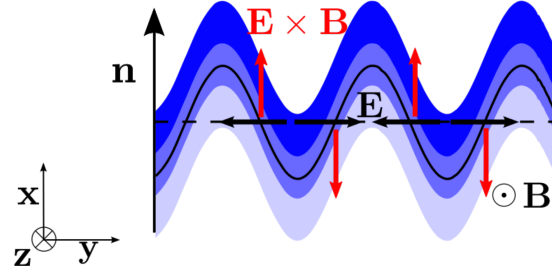


Figure 37: The cause of the drift wave.

Trapped particle modes

However if the particles are trapped then the electrons can't quickly stabilize, then the (slower) particle drift causes the ions and electrons to separate. This time to the transitionary region. This causes an E-field with the largest $E \times B$ force being at the extrema thus causing them to become larger (see figure 38). This is an trapped particle/electron mode and this causes the perturbation to grow and is thus *unstable*.

It should be noted that if the drifts are in the opposite direction, such that the resulting E-field would also be reversed, (for example on the inboard side of a Tokamak) that then the $E \times B$ force would work to reduce the perturbation, making it stable.

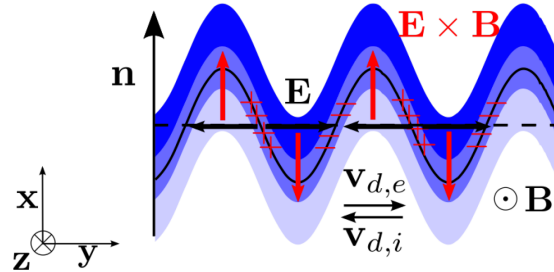
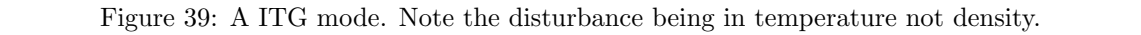


Figure 38: A trapped particle mode.

Ion temperature gradient mode (ITG)

Finally a disturbance in temperature can also cause problems. The drift-velocity also depends on temperature (the higher the faster). Thus a perturbation in temperature causes some ions to drift faster than others again causing a charge build-up between extrema, thus causing an $E \times B$ force mostly at the extrema. This causes perturbation growth and *instability* which is called the ion temperature gradient mode (see figure 39).

However, again like trapped particles modes, depending on the precise plasma conditions or B-field the particles might drift in opposite direction. Thus reducing the perturbation and causing stability.



- $\Delta\phi = 0$ or π (between extrema): neutral
- $-\pi > \Delta\phi > 0$ (opposite extrema): stable
- $0 > \Delta\phi > \pi$ (at extrema): unstable

Als niet eerder vermeld dan nu vermelden dat thresholds bestaan

Other exam question

Here are some other good/interesting questions from an exam, which I didn't know where to place.

Example: Pressure profile change due to sawtooth (exam April 2018 problem 3.h)

- (h) [6 pt] Suppose a tokamak plasma has a pressure profile $p(r) = p(0)(1 - r^2/a^2)$. First compute $p(0)$. A sawtooth instability occurs which ejects heat from the plasma core such that the pressure profile is flattened inside the magnetic surface where the instability is resonant. Compute the amount of thermal energy that is thus ejected from the core. Note that the thermal energy density is $\frac{3}{2}p$.

Solution: $p(0) = 2n_e k_B T = 2(10^{20})(5^3)(1.6022 \times 10^{-19}) = 1.6 \times 10^5 \text{ Pa} = 1.6 \text{ bar}$.

The sawtooth is resonant at $q = 1$ which is at minor radius $r_1 = (1.25)(0.19) = 0.24 \text{ m}$. The lost heat is $Q = \int \frac{3}{2} (p(r) - p(r_1)) d^3\mathbf{x} = \frac{3}{2} R_0 \int_0^{r_1} dr (2\pi r) (p(r) - p(r_1))$.

Filling in the numbers: $Q = 3\pi R_0 \int_0^{r_1} dr p(0) r (r_1^2 - r^2)/a^2 = \frac{3}{4}\pi R_0 p(0) r_1^4/a^2 = 4.8 \text{ kJ}$

Improve

Example: Perturbation in magnetic field (exam April 2018 problem 3.i)

- (i) [6 pt] This question does not refer to the tokamak. A homogeneous plasma has an equilibrium magnetic field $\mathbf{B}_0 = B_0 \mathbf{e}_z$. This plasma is perturbed with a velocity that is everywhere $\mathbf{u}(t, x, y, z) = u_1 \mathbf{e}_y \cos(\omega t - kz)$. Give the perturbed magnetic field.

Solution:

$$\begin{aligned} \frac{\partial \mathbf{B}_1}{\partial t} &= \nabla \times (\mathbf{u} \times \mathbf{B}_0) \\ &= \nabla \times (u_1 B_0 \mathbf{e}_x \cos(\omega t - kz)) \\ &= -kB_0 u_1 \mathbf{e}_y \sin(\omega t - kz) \end{aligned}$$

so that $\mathbf{B}_1 = (k/\omega) B_0 u_1 \mathbf{e}_y \cos(\omega t - kz)$.

Improve

Something about directions